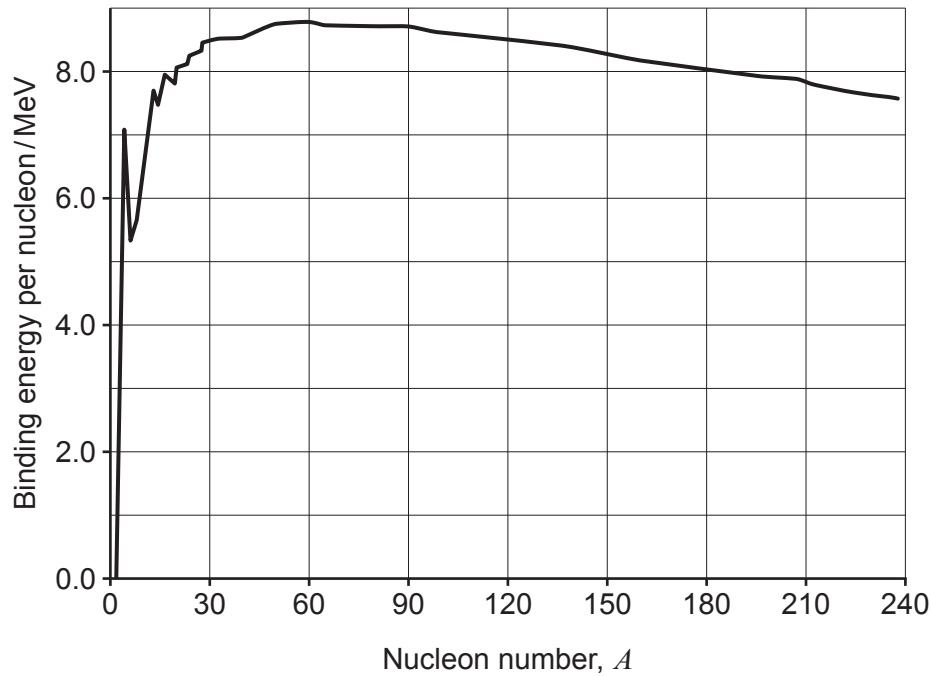


5. (a) Nuclear energy can be released by fusion and fission. Discuss these processes in terms of energy and the stability of the nuclei. Use the graph of the binding energy per nucleon in your answer. [4]



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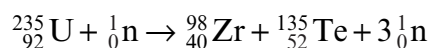
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- (b) One of the many possible fission reactions of uranium-235 is:



The masses are:

${}_{92}^{235}\text{U}$: 235.04393 u; ${}_{40}^{98}\text{Zr}$: 97.91273 u; ${}_{52}^{135}\text{Te}$: 134.91645 u, ${}_0^1\text{n}$: 1.00866 u
and 1 u = 931 MeV.

- (i) State what is meant by the Avogadro constant. [1]

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- (ii) Calculate the energy released by the fission of 0.0010 kg of uranium-235 in this reaction. Give your answer in J. [5]

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